Billel Helali

Senior Full Stack Engineer — 0→1 web & mobile, security by design

TypeScript · Next.js · React Native · Node.js · PostgreSQL · GDPR

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01 PROFILE

Senior full stack engineer (8 years since graduating), founder of Lumelia. I take products from idea to production on my own (architecture, web, mobile, backend, deployment, operations) and I have shipped them to real users: **Farouk.app** (2,000+ users, 100+ paying Pro), **ArabicPro** (iOS & Android, in TestFlight) and **EchoTrust** (B2B SaaS, first paying customers). Security and privacy are defaults, not a later phase. Background in regulated data: AML/KYC at an insurer, fraud detection on financial documents. I propose the architecture, challenge the scope and deliver.

02 PRODUCTS SHIPPED TO REAL USERS

Farouk.app — AI job-search assistant · B2C

LIVE · 2,000+ USERS

- **Built and run for a client, first commit to production:** CV parsing, job matching, automated applications by email and web forms, 6 languages, Stripe subscriptions.
- **Holds CVs and connected Gmail/Outlook mailboxes:** CASA AL1 (OWASP ASVS) pre-audit for Google's restricted scopes, row-level security per workspace, pgsodium-encrypted credentials (fail-closed), contact details redacted before any LLM call, GDPR erasure across database and file storage.
- **Run like a product, not a demo:** 450+ test files with critical suites gating every PR, health checks every 30 min, staging-to-production promotion, additive-only migrations, post-merge guard that reverts on regression.

STACK Next.js 14 · TypeScript · FastAPI · Python workers · Supabase/PostgreSQL · Stripe · Vercel · Railway

ArabicPro — Arabic-learning mobile app · B2C · iOS & Android

TESTFLIGHT

- **Built for a client with Expo SDK 54 / React Native 0.81 / TypeScript:** audio lessons, Sign in with Apple, RevenueCat subscriptions, push notifications, EAS builds and OTA updates per channel.
- **Children's data by design:** parental consent split between general use and voice recordings, HMAC-signed expiring links carrying no personal data, in-app account deletion, deny-by-default rules on Data Connect (PostgreSQL).
- **Next.js 16 admin back-office** behind signed session cookies and an admin claim; 570+ test files, CI with lint, type-check, unit and E2E; Sentry and Crashlytics.

STACK Expo · React Native · TypeScript · Firebase Auth · Data Connect/PostgreSQL · RevenueCat · Next.js 16

EchoTrust — Review management for multi-location businesses · B2B SaaS

LIVE · FIRST CUSTOMERS

- **Next.js 16 / React 19 on Vercel:** Clerk auth, PostgreSQL with Drizzle, Stripe with signed webhooks, rate limiting, Google Business Profile OAuth, LLM-drafted replies published only after human approval.
- **Secrets handled properly:** OAuth tokens encrypted with AES-256-GCM (HKDF-derived keys), API keys stored as salted hashes with constant-time checks, MCP server with workspace-scoped keys and dry-run on destructive writes, secret scanning and SHA-pinned CI actions.

STACK Next.js 16 · React 19 · TypeScript · PostgreSQL · Drizzle · Clerk · Stripe · Vercel · Playwright

03 SENSITIVE DATA I HAVE PROTECTED

Regulated finance · Seyna
AML/KYC (LCB-FT) obligations turned into deterministic, **auditable** rules with compliance; 5M+ registry records a month, traceable.

Financial documents · Doksima
Payslips and bank statements read by **local OCR models shipped in the container**; retention-based purge built in; every verdict cites its evidence.

Mailboxes & CVs · Farouk.app
Restricted Gmail scopes under a **CASA / OWASP ASVS** pre-audit, PII redacted before LLMs, **GDPR erasure** down to stored files.

Children & voice · ArabicPro
Parental consent covering voice recordings, signed links with no personal data, in-app deletion, deny-by-default access.

Secrets, tokens & auth
AES-256-GCM, pgsodium, HMAC, salted hashed API keys, constant-time checks, fail-closed encryption, OAuth 2.0, session cookies.

Privacy-safe analytics & AI
Aggregates shown only for groups of **10+ people**; prompt-injection hardening: risky actions gated in code, never by the model.

04 EXPERIENCE

Sep 2025 – now
Rennes · remote

Founder & Lead Engineer — Lumelia

Product studio. Sole engineer on Farouk.app, ArabicPro and EchoTrust (above): scoping with the client, architecture, web, mobile, backend, deployment and operations.

- **Doksima (R&D proof of concept):** fraud detection on French supporting documents. Models locate the evidence, a deterministic engine decides (to-the-cent arithmetic, IBAN/SIRET/VAT, offline 2D-Doc signature checks) and abstains below threshold; ~2,900 automated tests.
- **Maison Autret:** mapped the real sales and marketing process, audited HubSpot and Make, built a React/TypeScript analytics dashboard; a 21-day sprint became an ongoing engagement.

STACK TypeScript · Next.js · React Native · Python · PostgreSQL · Vercel · AWS/GCP

Oct 2022 – Apr 2025
Paris · remote

Senior Software Engineer — Backend & Cloud — Seyna (insurtech)

- **Turned AML/KYC (LCB-FT) obligations into deterministic, auditable system rules** with the compliance team, shipped as serverless services on AWS.
- **Designed and ran an ingestion pipeline for 5M+ company-registry records a month:** data-quality rules, deduplication, retries, cross-source reconciliation.
- **Built production APIs in Node.js and Python** (SQL and NoSQL) and their partner contracts; event-driven workflows on EventBridge, SNS, SQS, SES. **Led the AI workstream from 2024** (agents in underwriting).

STACK AWS · EventBridge · SQS · SNS · SES · Node.js · Python · SQL/NoSQL

Mar 2021 – Sep 2022
Paris

Senior Software Engineer & Tech Lead — Arcane

SmartFeeds, product-feed platform used by Leroy Merlin, Carrefour, Decathlon, Sézane and L'Occitane.

- **Led a React/TypeScript team for 18 months:** scope trade-offs with product, releases cut into deployable increments, code reviews; **introduced testing standards** that cut critical defects in production.
- **Designed containerised Python/Flask microservices** and the segmentation rules (margin, profitability, competitiveness) that drove real ad spend on Google Ads, Meta and Pinterest.

STACK React · TypeScript · Python/Flask · Docker · Microservices

2017 – 2021
Rennes

Software Engineer — Freelance · Harmonic Inc.

- **Aviwest** (mission-critical video SaaS), **IFSTTAR** (3D research-data viewer), **PettyWell** (product delivery, third-party integrations). **Harmonic Inc.** internship: ML video-quality models, team across France, Israel and the US.

05 TECH STACK

FRONTEND & MOBILE	TypeScript, React 19, Next.js 14–16 (App Router), React Native & Expo (EAS Build, OTA updates, RevenueCat), Tailwind CSS, i18n
BACKEND	Node.js, NestJS, Python, FastAPI, REST APIs, webhooks, background workers and queues (BullMQ, SQS)
DATA	PostgreSQL (Supabase, RDS, Firebase Data Connect), Drizzle, row-level security, migrations, Redis, MongoDB
SECURITY & PRIVACY	OAuth 2.0, Clerk, Firebase Auth, Sign in with Apple, AES-256-GCM, pgsodium, HMAC, OWASP ASVS, GDPR erasure & minimisation
CLOUD & DELIVERY	Vercel, AWS, GCP/Firebase, Railway, Docker, Kubernetes/EKS, GitHub Actions, GitLab CI, staged releases
QUALITY & OPS	Jest, Vitest, Pytest, Playwright, Sentry, Crashlytics, OpenTelemetry, Prometheus/Grafana
AI IN PRODUCT	LLM integration (OpenAI, Anthropic, Mistral), agents and MCP, structured outputs, guardrails, evaluation

06 EDUCATION & LANGUAGES

ENGINEERING DEGREE	**ESIR, Engineering School of the University of Rennes** — Computer Science · 2014 – 2018 · Master's level (EQF 7)
LANGUAGES	**English** — C1, daily working language · **French** — native